File format version 5

Format Specification

Copyright © 2009 Marcus Geelnard

Contents

1 Introduction **2**

2 File format specification **3**

2.1 Data formats 3

2.2 Header 3

2.3 Body data 4

2.3.1 RAW 4

2.3.2 MG1 4

2.3.3 MG2 5

Chapter 1

Introduction

This document describes version 5 of the OpenCTM file format.

Chapter 2

File format specification

The structure of a OpenCTM file is as follows:

[Header]
[Body data]

Each part of the file is described in the following sections.

2.1 Data formats

All integer fields are stored in 32-bit little endian format (least significant byte first).

All floating point fields are stored in 32-bit binary IEEE 754 format (little endian).

All strings are stored as a 32-bit integer string length (number of bytes) followed by a UTF-8 format string (there is no zero termination and no BOM).

2.2 Header

The file must start with a header, which looks as follows:

Offset	Type	Description
0	Integer	Magic identifier (0x4d54434f, or ”OCTM” when read as ASCII).
4	Integer	File format version (0x00000005 = version 5).
8	Integer	Compression method, which can be one of the following: 0x00574152 - Use the RAW compression method. 0x0031474d - Use the MG1 compression method. 0x0032474d - Use the MG2 compression method.
12	Integer	Vertex count.
16	Integer	Triangle count.
20	Integer	UV map count.
24	Integer	Attribute map count.
28	Integer	Boolean flags, or:ed together: 0x00000001 - The file contains per-vertex normals.
32	String	File comment.

The length of the file header is varibale, since it contains fields of variable length.

2.3 Body data

The body data follows immediately after the file header. Its file offset is dictated by the length of the file header.

The format of the body data is specific for each compression method, which is defined by the ”Compression method” field in the header.

The body data contains the vertex, index, normal, UV map and attribute map data, usually in a compressed form.

2.3.1 RAW

To be written...

Please read the source code file compressRAW.c for more details on the body format when the RAW compression method is used.

2.3.2 MG1

To be written...

Please read the source code file compressMG1.c for more details on the body format when the MG1 compression method is used.

2.3.3 MG2

To be written...

Please read the source code file compressMG2.c for more details on the body format when the MG2 compression method is used.